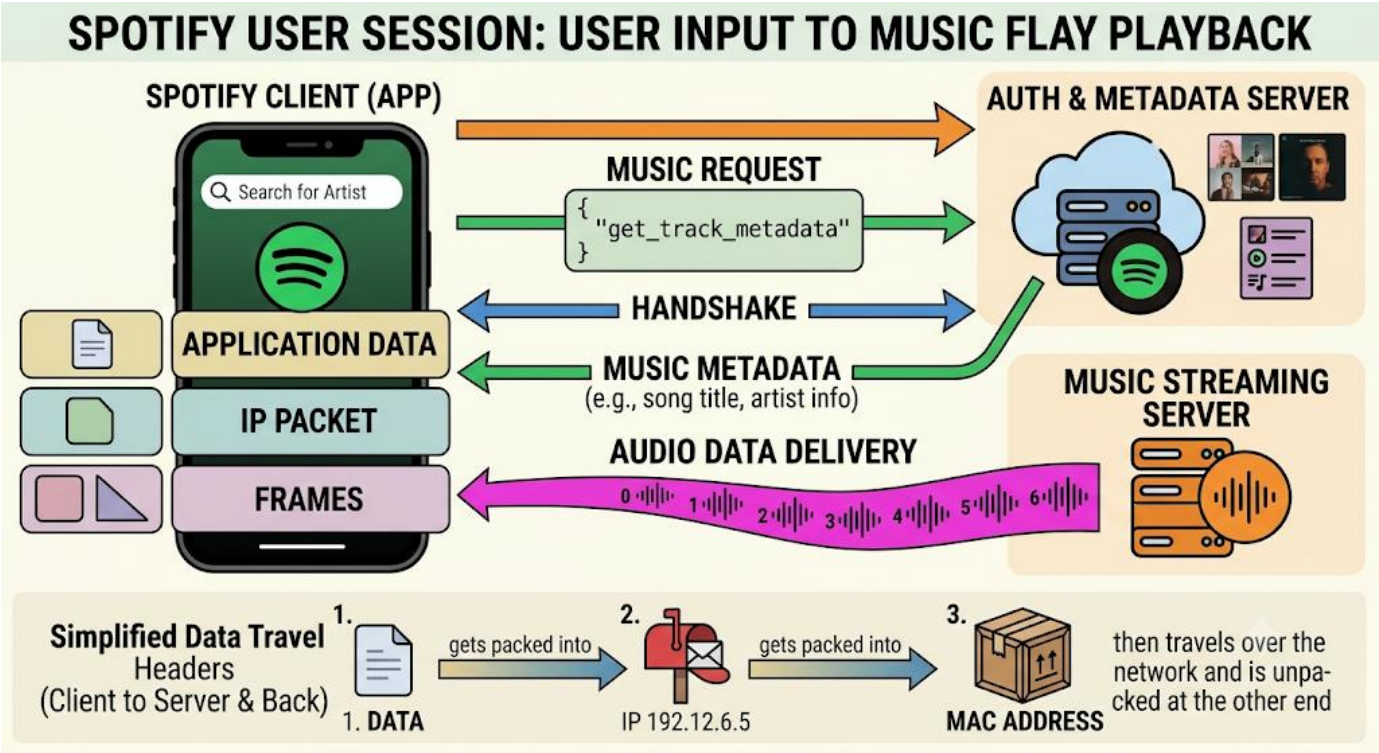
## MODULE 1 ASSIGNMENT – COMPUTER NETWORKS (KTU 2024 SCHEME)

**Website Chosen: Spotify.com**

### 1. TCP/IP Protocol Stack




The TCP/IP model has four layers: Application, Transport, Internet and Network Access. The Application layer provides services such as HTTPS and DNS. The Transport layer uses TCP to establish reliable communication between the Spotify client and server. The Internet layer uses IP to route packets through different networks until they reach the destination. The Network Access layer transmits frames through Ethernet or Wi-Fi. Every request and response passes through these layers before reaching the user. Diagram:
Application → Transport (TCP) → Internet (IP) → Network Access.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

### 2. Role of DNS

When the user types spotify.com into a browser, a DNS resolver contacts DNS servers to obtain the corresponding IP address. Without DNS, users would need to remember numerical IP addresses instead of simple domain names. Spotify also uses

geographically distributed DNS services so users are connected to the nearest server, improving speed and reducing latency.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

### 3. HTTP/HTTPS

Spotify mainly uses HTTPS to provide secure communication. The client sends an HTTPS GET request for webpages, APIs and account information. The server authenticates the request and returns encrypted responses containing HTML, CSS, JavaScript, images and streaming metadata. Encryption protects user credentials and prevents attackers from reading transmitted data.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

### 4. TCP Reliability

TCP begins communication with a three-way handshake. It divides data into segments, assigns sequence numbers, waits for acknowledgements and retransmits missing packets. Flow control prevents the receiver from becoming overloaded, while congestion control reduces network congestion. These features ensure songs and webpages are delivered accurately and in order.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

### 5. Client–Server Architecture

Spotify follows the Client–Server model because users access centralized Spotify servers using web browsers or mobile applications. Clients request songs, playlists, podcasts and account details, while Spotify servers process these requests and return the required information. This architecture simplifies management and supports millions of simultaneous users.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

## 6. HTTP vs FTP

HTTP is designed for browsing web pages and web applications, while FTP is designed for transferring files between computers. HTTP uses ports 80 and 443, whereas FTP commonly uses port 21. HTTP is stateless, while FTP maintains a session during file transfer. HTTP transfers webpages and multimedia resources, whereas FTP uploads and downloads complete files.

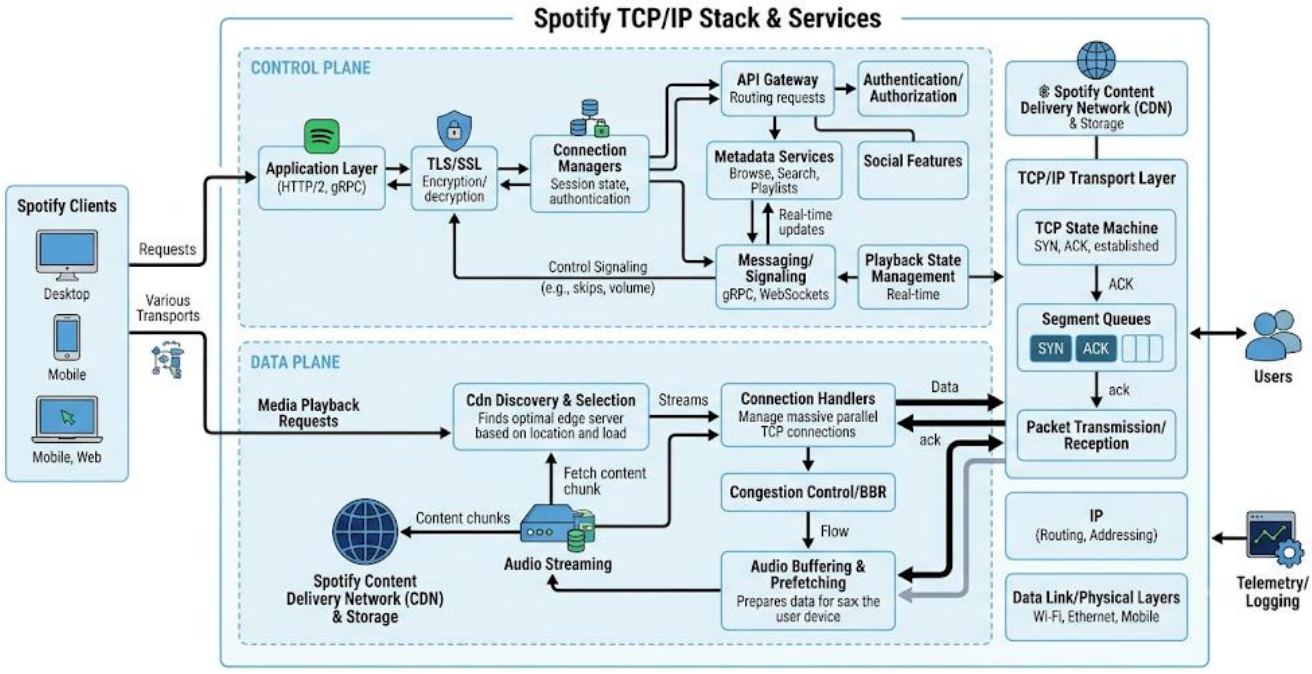
Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

## 7. Cookies

Cookies are small text files stored by the browser. Spotify uses cookies to remember login sessions, preferred language, playback settings and recently played content. Cookies improve user experience by reducing repeated logins and providing personalized recommendations based on previous activity.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.

## 8. Communication Flow



User enters spotify.com → DNS resolves the domain → Browser receives IP address → TCP connection is established → HTTPS request is sent → Spotify server processes the request → HTTPS response is returned → Browser displays the webpage or starts streaming audio. This complete process happens within a few seconds.

## 9. Conclusion

Application-layer protocols play a vital role in everyday Internet communication. DNS identifies the destination server, TCP guarantees reliable communication and HTTPS securely transfers webpages and streaming information. Together these protocols allow Spotify to deliver fast, secure and uninterrupted music streaming. They improve reliability, protect user privacy and ensure that millions of users around the world can access online content efficiently at any time.

Additional Note: In large content delivery platforms such as Spotify, caching, load balancing and Content Delivery Networks (CDNs) are also used to improve performance, reduce latency and provide uninterrupted service during periods of high traffic.